

Employee Turnover Analytics Report

Business Problem

Portobello Tech wants to predict employee turnover and identify the main factors that influence attrition so HR can take early retention action.

Dataset Used

The HR dataset contains 14999 records and 10 columns. After quality review, 3008 duplicate rows were removed, leaving 11991 unique employee records for analysis.

Project Scope

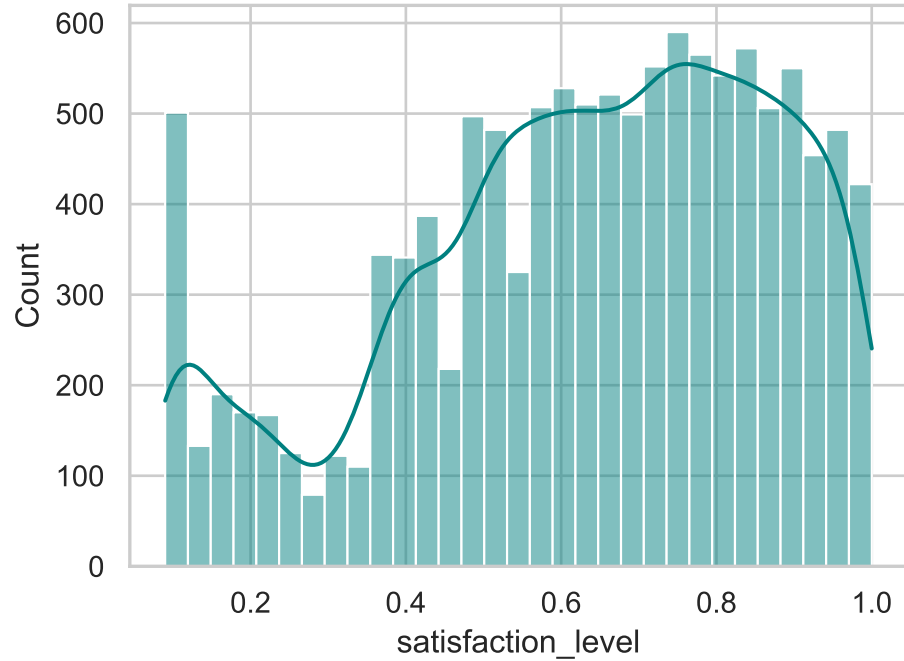
The work includes data quality checks, exploratory analysis, K-means clustering for employees who left, SMOTE-based class balancing, 5-fold cross-validation for three models, model comparison, and retention strategy recommendations.

Deliverables

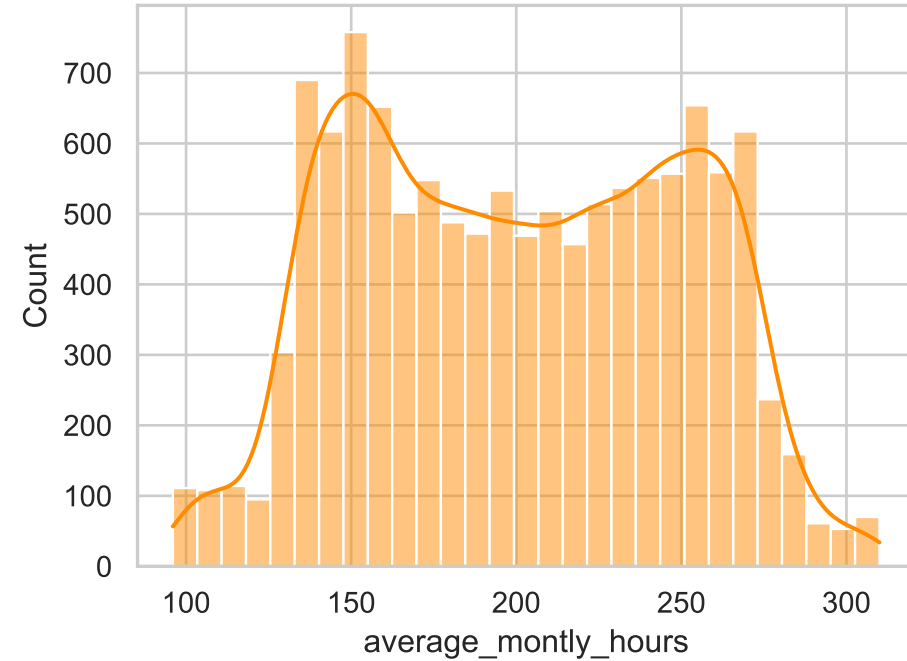
Two deliverables were prepared: an executed Jupyter notebook for code and outputs, and this PDF write-up for submission-ready documentation.

Exploratory Data Analysis

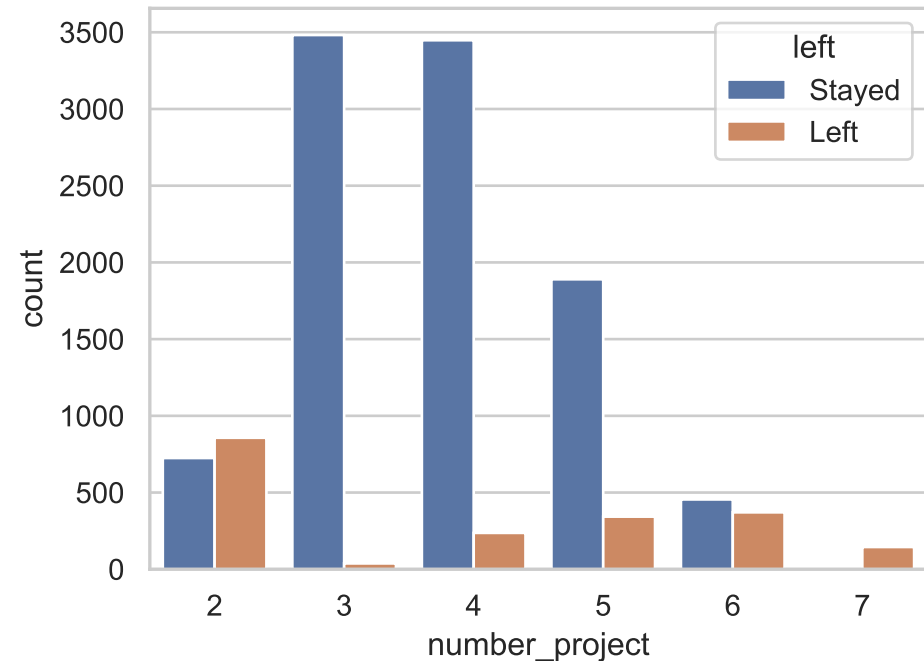
Satisfaction Level Distribution



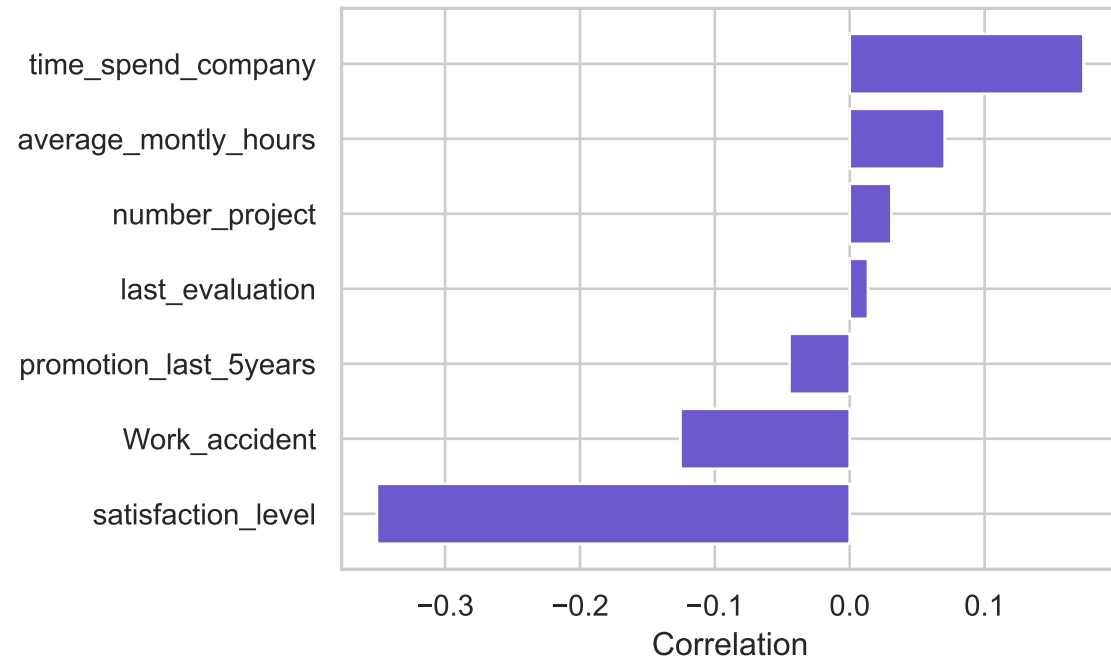
Average Monthly Hours Distribution



Project Count by Turnover Status



Correlation with Turnover



EDA Findings

Data Quality

No missing values were found in the dataset. The main quality issue was duplication: 3008 rows were duplicates and were removed before modeling.

Key Drivers

The strongest turnover signal is low satisfaction level. Longer time spent in the company and heavier monthly workload also increase attrition risk, while work accidents and promotions show negative association with turnover.

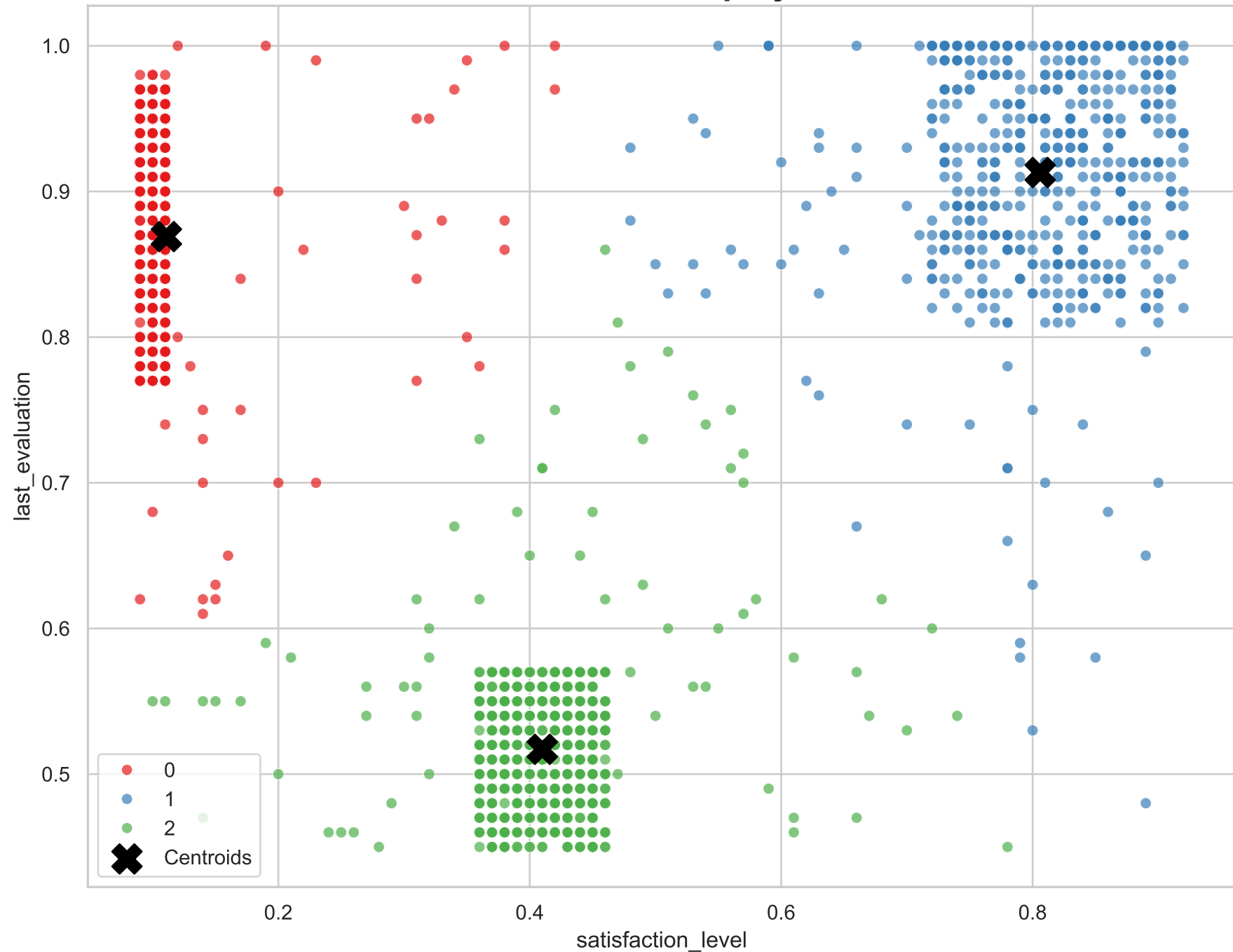
Project Workload Insight

Employees with 2 projects had a turnover rate of 54.17%, and those with 7 projects had a turnover rate of 100.00%. In contrast, employees with 3 or 4 projects were much more stable.

Salary Insight

Employees in the low-salary band had the highest turnover rate at 20.45%, compared with 4.85% in the high-salary band.

K-Means Clusters for Employees Who Left



Cluster Summary

Clusters were built only on employees who left, using satisfaction and last evaluation.

Cluster	Avg Satisfaction	Avg Evaluation	Employees
0.0	0.111	0.869	534.0
1.0	0.806	0.913	555.0
2.0	0.41	0.517	902.0

Clustering Interpretation

Cluster 0

Very low satisfaction with high evaluation. These employees appear productive but deeply dissatisfied, which is a strong indicator of likely burnout, poor manager fit, or reward mismatch.

Cluster 1

High satisfaction and high evaluation, yet these employees still left. This suggests external opportunities, compensation expectations, or career progression limits may have driven attrition.

Cluster 2

Moderate satisfaction and moderate evaluation. This group reflects disengaged or average-performing employees who may have lacked strong attachment to the organization.

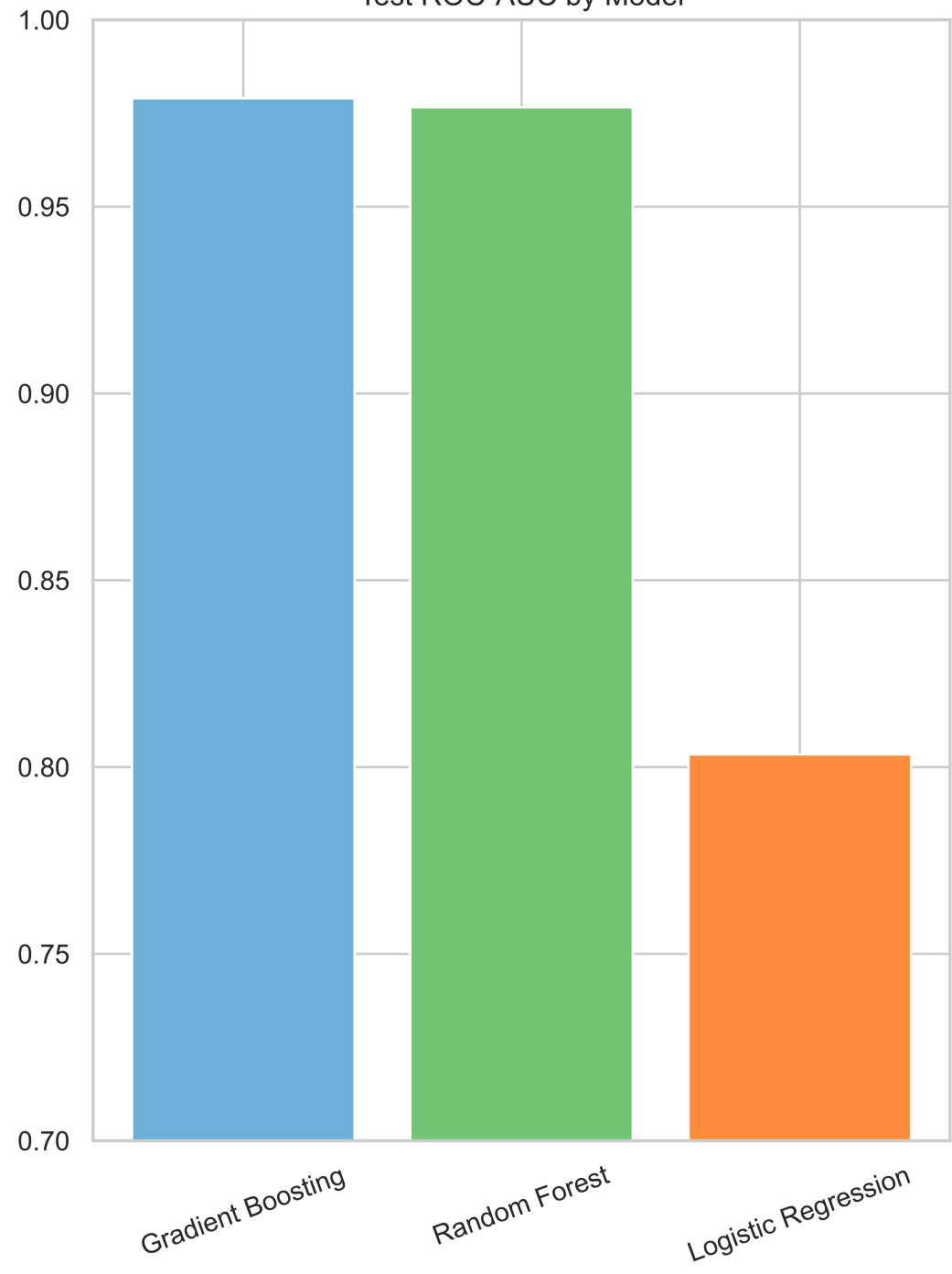
Model Comparison

All models used SMOTE on training folds and were evaluated with 5-fold cross-validation.

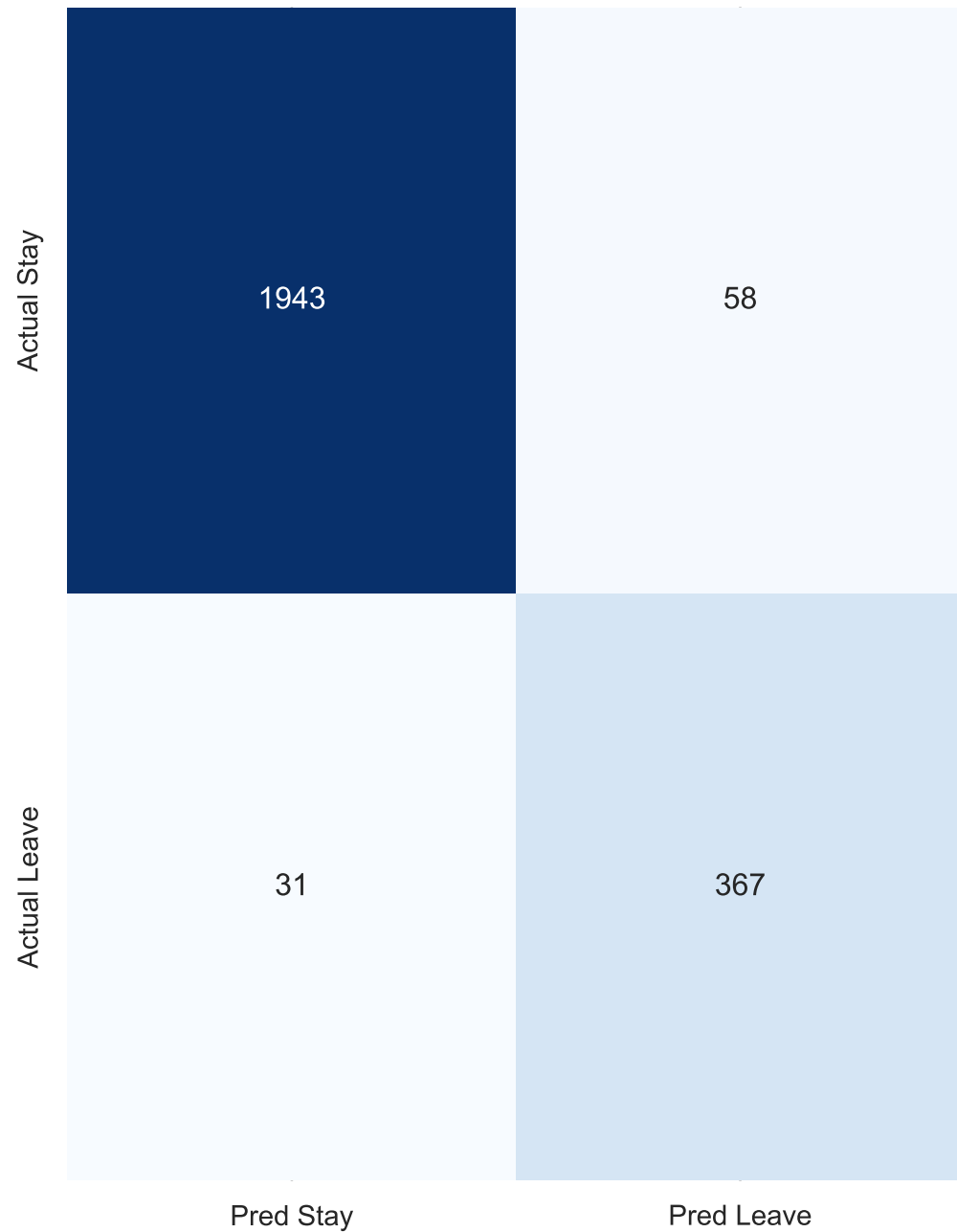
Model	CV ROC-AUC	Test ROC-AUC	Precision	Recall	F1-Score
Gradient Boosting	0.985	0.979	0.8635	0.9221	0.8919
Random Forest	0.9817	0.9766	0.9474	0.9045	0.9254
Logistic Regression	0.8096	0.8034	0.4267	0.6658	0.5201

Model Performance

Test ROC-AUC by Model



Confusion Matrix: Gradient Boosting



Best Model and Metric Justification

Selected Model

Gradient Boosting was selected as the best model because it achieved the highest test ROC-AUC of 0.9790 while also maintaining strong recall and balanced overall performance.

Why ROC-AUC

ROC-AUC is appropriate for comparing models because it evaluates ranking quality across thresholds rather than at a single cutoff point.

Why Recall Matters

For attrition prediction, recall is especially important because a false negative means the business failed to identify an employee who is actually likely to leave. Missing those employees weakens the value of the retention program.

Best Model Test Metrics

Precision = 0.8635, Recall = 0.9221, and F1-score = 0.8919. The confusion matrix for the best model was $\begin{bmatrix} 1943 & 58 \\ 31 & 367 \end{bmatrix}$.

Risk Zone Summary

Risk zones were created from predicted turnover probabilities on the test set.

Risk Zone	Employees	Avg Turnover Probability (%)	Actual Turnover Rate (%)
Safe Zone (Green)	1840	4.29	1.03
Low-Risk Zone (Yellow)	155	33.86	9.03
Medium-Risk Zone (Orange)	68	77.64	54.41
High-Risk Zone (Red)	336	96.67	97.62

Retention Recommendations

Safe Zone (Green)

Continue normal engagement through recognition, regular manager check-ins, and growth conversations. These employees show low predicted turnover probability.

Low-Risk Zone (Yellow)

Monitor workload and job satisfaction. Offer career development discussions and manager feedback sessions before risk escalates.

Medium-Risk Zone (Orange)

Use targeted intervention: role review, project rebalancing, skill development, and compensation benchmarking where needed.

High-Risk Zone (Red)

Immediate action is recommended. HR and managers should conduct one-on-one retention meetings, review compensation, evaluate internal mobility options, and address dissatisfaction or overload quickly.

Conclusion

Overall Outcome

Employee turnover can be predicted effectively using historical HR data. Satisfaction, tenure, workload, and project pressure are the most important attrition signals.

Business Value

The model can help HR prioritize intervention efforts and allocate retention resources to employees with the highest probability of leaving.

Final Recommendation

Use the best model as a screening tool alongside HR judgment, then apply different retention strategies by risk zone rather than using a single intervention for all employees.